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ASSIGNMENT No: 15

TITLE: Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Theory

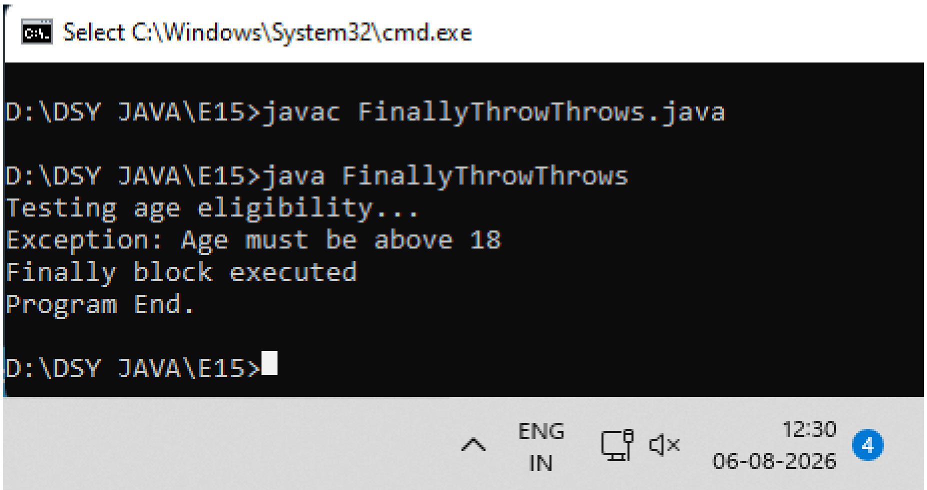
The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

Code

```
import java.io.*;

public class FinallyThrowThrows{

   static void checkAge(int age)throws IOException{
      if (age < 18){
         throw new IOException("Age must be above 18");
      }
      System.out.println("Eligible for Voting.");
   }

   public static void main(String a[]){

      try {
         System.out.println("Testing age eligibility...");
         checkAge(15);

      } catch (IOException e) {

         System.out.println("Exception: " + e.getMessage());
```

```
        } finally {

            System.out.println("Finally block executed");
        }

        System.out.println("Program End.");
    }
}
```

Output



Exercise

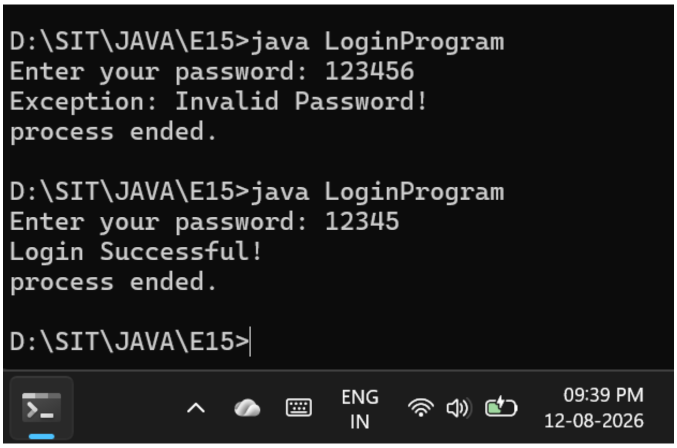
1. Create a Login program that throws an exception for invalid password and uses finally block.

Code

```
import java.io.*;
import java.util.Scanner;

public class LoginProgram {

    static void login(String password) throws Exception {
        if (!password.equals("12345")) {
            throw new Exception("Invalid Password!");
        }
        System.out.println("Login Successful!");
    }
```

```
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter your password: ");
        String userInput = scanner.nextLine();

        try {
            login(userInput);

        } catch (Exception e) {
            System.out.println("Exception: " + e.getMessage());

        } finally {
            System.out.println("process ended.");
            scanner.close();
        }
    }
}
```

Output



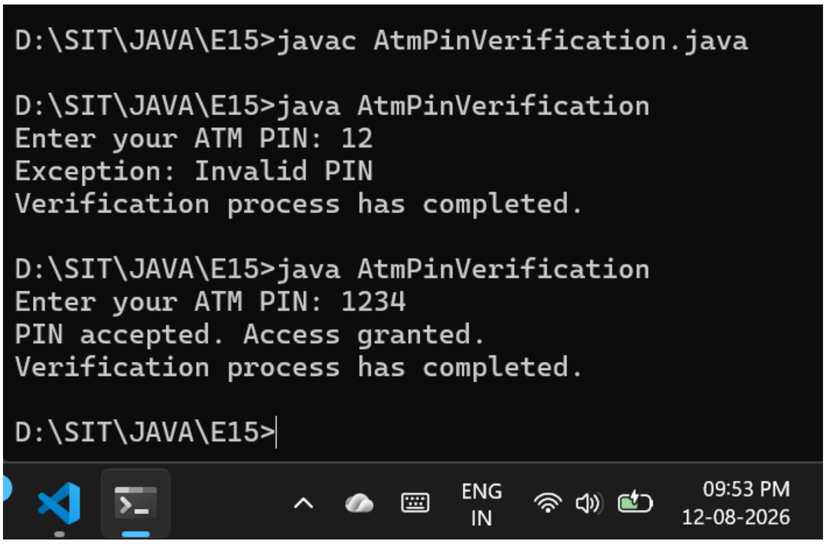
2. Create an **ATM PIN Verification** program that throws an exception for an invalid PIN entered by the user and uses a **finally** block to display a message indicating that the verification process has completed.

Code

```
import java.io.*;
import java.util.Scanner;

public class AtmPinVerification {
```

```
    static void verifyPin(int pin) throws Exception {
        int correctPin = 1234;

        if (pin != correctPin) {
            throw new Exception("Invalid PIN");
        }
        System.out.println("PIN accepted. Access granted.");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your ATM PIN: ");

        try {
            int enteredPin = sc.nextInt();
            verifyPin(enteredPin);

        } catch (Exception e) {
            System.out.println("Exception: " + e.getMessage());

        } finally {
            System.out.println("Verification process has completed.");
            sc.close();
        }
    }
}
```

Output



```
D:\SIT\JAVA\E15>javac AtmPinVerification.java

D:\SIT\JAVA\E15>java AtmPinVerification
Enter your ATM PIN: 12
Exception: Invalid PIN
Verification process has completed.

D:\SIT\JAVA\E15>java AtmPinVerification
Enter your ATM PIN: 1234
PIN accepted. Access granted.
Verification process has completed.

D:\SIT\JAVA\E15>
```